Seat No.	

Set P

B.Sc. (E.C.S.) (Semester - II) (New) (NEP CBCS) Examination: March/April – 2025
Discrete Mathematics (G06-0204)

Day & Date: Saturday, 17-05-2025 Max. Marks: 30
Time: 12:00 PM To 01:30 PM

Instructions: 1) All questions are compulsory.
2) Figures to the right indicate full marks.

Q.1 Choose correct alternative for the following. 06

1) The relation is called _____ relation if a R b implies b R a.
a) Symmetric b) Reflexive
c) Transitive d) Equivalence

2) A relation which is Symmetric, Reflexive and Transitive is called _____.
a) Symmetric b) Reflexive
c) Transitive d) Equivalence

3) A function which is Injective and surjective is called _____.
a) one-one b) on-to
c) Bijective d) none of these

4) let $f(x) = 2x + 6$, find the value of function at $x = -2$ _____.
a) 2 b) -2
c) 6 d) 4

5) The relation is called _____ relation if a R a for all a belongs to A.
a) Symmetric b) Reflexive
c) Transitive d) Equivalence

6) A set having only one element is called _____ set.
a) singleton b) empty
c) infinite d) none of these

Q.2 Answer any three of the following. 06

a) Define Function.
b) Define Equivalence relation.
c) State inclusive-exclusive principle for two sets.
d) Let the set $A = \{1,3,5,7\}$ and $B = \{1,2,3,4\}$ then find set AxB.
e) Define homogeneous recurrence relation.

Q.3 Answer any two of the following. **06**

a) Write note on matrix representation of a relation.

b) Define Partial ordering relation.

c) If $F(x) = 2x^3 + 6x$ then find

i) $f(2)$

ii) $f(4)$

Q.4 Answer any two of the following. **06**

a) Let the function $F: R \rightarrow R$ Define as, $F(x) = 4x + 3$. show that f is Bijective.

b) for the given relation

$R = \{(1,1), (1,2), (2,1), (2,2), (2,3), (2,4), (3,4), (4,1)\}$ on $A = \{1,2,3,4\}$

i) Matrix of relation R

ii) Diagraph and

iii) In degree and Out degree of the elements of A

c) Solve the recurrence relation $a_r + 7a_{r-1} + 12a_{r-2} = 0$

Q.5 Answer any one of the following. **06**

a) Solve the recurrence relation $a_r{}^{-} - 7a_{r-1} + 6a_{r-2} = 0$ with initial conditions $a_0 = 15$ and $a1 = 5$.

b) Let the set $A = \{a, b, c, d, f\}$ Define relation R on set A is

$A = \{(a, a), (a, c), (a, f), (b, c), (b, d), (b, f), (c, a), (c, d), (c, f), (d, a), (d, c),$ $(f, c), (f, f)\}$. Find transitive closure R^* by using wars halls algorithm.